1.Introduction:

This report presents an analysis focused on predicting hypertension prevalence using historical data. Utilizing the comprehensive World Bank dataset, which provides prevalence rates across various countries and years, the primary objective is to develop a predictive model for future hypertension trends.

Our approach involves employing Linear Regression as the core predictive model, which is a powerful statistical method used to model the relationship between a dependent variable (hypertension prevalence) and one or more independent variables (such as age, gender, income, etc.). Linear regression works by fitting a linear equation to observed data, where the model coefficients represent the strength and direction of relationships between predictors and the target variable.

To enhance the model's performance and interpretability, we will first apply filter methods-based feature selection techniques. These methods will help us identify the most relevant historical prevalence data points, ensuring that our model is built upon the most informative features while mitigating issues like multicollinearity and noise. This structured methodology aims to deliver robust predictions and a clearer understanding of the factors influencing hypertension prevalence over time.

2.Dataset Description:

This analysis utilizes the World Bank's '**Prevalence of hypertension, ages 30-79 (%)'** dataset, sourced from the World Bank DataBank portal. The data was loaded from a CSV file named WB_HCP_HYPERTENSION_WIDEF.csv, which was accessed from Google Drive.

The dataset contains 576 rows and 50 columns, covering the time span from 1990 to 2019. Key columns include:

'REF_AREA_LABEL': Represents the country name (e.g., Afghanistan, United States).
'SEX_LABEL': Indicates the demographic group (Female, Male, or Total).
Yearly prevalence columns: These columns are named after the years, ranging from '1990' to '2019', and contain the hypertension prevalence percentages for that specific year.

3.Methodology:

3.1 Data Loading and Initial Inspection

Before applying Linear Regression, several important preprocessing steps were carried out to prepare the data. First, the dataset was loaded into a pandas DataFrame, and initial inspections using commands like df.head() and df.info() were performed to understand its structure and ensure the absence of missing values and duplicates. Outliers were examined through box plots for each year’s hypertension prevalence, confirming that there were no extreme values that required removal. Duplicate checks further confirmed the uniqueness of

the data. To make the dataset suitable for model training, feature selection was applied to reduce dimensionality, with a focus on selecting relevant years based on correlation analysis. Once the data was cleaned, it was partitioned into training (60%), validation (20%), and test (20%) sets using the train_test_split() function. Standardization of the features was then applied using StandardScaler, fitted only to the training data to prevent data leakage. This step ensured that all features had a mean of 0 and a standard deviation of 1, making them suitable for linear regression and other machine learning algorithms. With the data preprocessed, the next step is to apply Linear Regression and feature selection techniques.

3.2 Linear Regression with Filter Methods

In this step, **Linear Regression** was applied to predict the hypertension prevalence for 2015 based on historical data from earlier years. The primary goal was to establish a linear relationship between the target variable (2015 hypertension prevalence) and the features representing the years 1990, 1995, 2000, 2005, and 2010. Prior to fitting the Linear Regression model, **filter methods** were applied to select the most relevant features, ensuring that the model would focus on the most informative data points. After applying these methods, the following years were selected for further analysis:

- **1990**
- **1995**
- **2000**
- **2005**
- **2010**

These years were chosen due to their strong correlations with the 2015 hypertension prevalence and their ability to capture long-term trends. The next step was to perform feature selection using various statistical methods, followed by fitting the Linear Regression model using the selected features.

3.3 Filter Methods and Feature Selection

3.3.1 Pearson Correlation Coefficient-

The **Pearson Correlation Coefficient** was used to measure the linear relationship between each selected feature (the historical years) and the target variable (2015 hypertension prevalence). The purpose was to identify which features had the strongest linear correlation with the target variable. The following code snippet was used to calculate the correlation and identify the most significant feature:

```
correlations = combined_df.corr()['target'].drop('target')
sorted_correlations = correlations.abs().sort_values(ascending=False)
```

This method calculated the Pearson correlation between the features and the target variable, and the results were sorted by absolute correlation. The features with the highest correlations were identified, with years like 2010 and 2005 showing particularly strong correlations to the 2015 prevalence, making them key predictors for the model.

And the output is:

```
Top correlated features with the target variable (ranked by absolute correlation):
2010    0.966613
2005    0.891182
2000    0.829139
1995    0.778188
1990    0.722293
Name: target, dtype: float64
```

As seen in the output, the years closer to 2015, particularly 2010, show the highest correlation with the target variable, indicating that these years are the most predictive for hypertension prevalence in 2015. The strong correlations observed for these years suggest that they provide the most significant insights into the trends of hypertension prevalence over time, making them the key features for the Linear Regression model.

3.3.2 Variance Threshold

The Variance Threshold method was applied to eliminate features with low variability, ensuring that only features with sufficient information would be retained for further analysis. With a threshold set to 0.05, all five historical years ('1990', '1995', '2000', '2005', and '2010') were retained, as none of them showed low variance.

```
selector = VarianceThreshold(threshold=0.05)
selector.fit(X_train_scaled)

# Get names of selected features
features_after_variance_threshold = [feature_years_for_split[i] for i, selected in
enumerate(selector.get_support()) if selected]

print("Selected features after VarianceThreshold:")
print(features_after_variance_threshold)
```

The output confirmed that all features exhibited adequate variability, meaning none were near-constant and thus were considered relevant for the model.

```
Original number of features: 5
Number of features after VarianceThreshold: 5
Selected features after VarianceThreshold:
['1990', '1995', '2000', '2005', '2010']
```

3.3.3 Mutual Information (Information Gain)

Mutual Information was used to evaluate the dependency between each feature and the target variable (2015 hypertension prevalence). This method measures the amount of information gained about the target variable by knowing the feature. The **2010** feature was found to have the highest information gain, followed by **2005**, **2000**, **1990**, and **1995**.

```
mi_scores = mutual_info_regression(X_train_scaled, y_train, random_state=42)
mi_series = pd.Series(mi_scores, index=feature_years_for_split)

sorted_mi_features = mi_series.sort_values(ascending=False)

print("Features ranked by Mutual Information (Information Gain) with the target:")
print(sorted_mi_features)
```

The output confirmed that all five features provided valuable information, with the strongest information gain observed for the years closer to 2015.

```
Features ranked by Mutual Information (Information Gain) with the target:
2010    1.455366
2005    0.880632
2000    0.734215
1990    0.675584
1995    0.657225
dtype: float64
```

3.3.4 Mean Absolute Difference (MAD)

The **Mean Absolute Difference (MAD)** was calculated to assess the variability of each feature in relation to its mean. The **2010** feature exhibited the highest MAD, indicating the greatest dispersion from its mean, followed by **2005**, **2000**, **1995**, and **1990**. This suggests that the features with higher MAD values contribute more diverse information to the model.

```
mad_values = (X_train_df_mad - X_train_df_mad.mean()).abs().mean()

sorted_mad_features = mad_values.sort_values(ascending=False)
print("Features ranked by Mean Absolute Difference (MAD) in X_train_scaled:")
print(sorted_mad_features)
```

The output confirmed that the **2010** feature had the most significant dispersion, making it a valuable predictor for hypertension prevalence in 2015, followed closely by other key years.

```
Features ranked by Mean Absolute Difference (MAD) in X_train_scaled:
2010    0.800335
2005    0.777060
2000    0.758848
1995    0.737341
1990    0.720036
dtype: float64
```

The Final selected features for linear regression model:
['1990', '1995', '2000', '2005', '2010']

The final selected years for the Linear Regression model were determined based on the results of multiple feature selection techniques, including Pearson Correlation Coefficient, Variance Threshold, and Mutual Information. These years—**1990**, **1995**, **2000**, **2005**, and **2010**—demonstrated strong relationships with the target variable, 2015 hypertension prevalence, based on their high correlation, significant information gain, and sufficient variability. These features were deemed most relevant for predicting hypertension prevalence in 2015 and were ultimately selected for use in the Linear Regression model, ensuring a robust and meaningful predictive model.

3.4 Predicting y-values

Before evaluating the model, we first predicted the hypertension prevalence values for the validation set using the trained Linear Regression model. This gives us the model's predictions on unseen data, which will later be used to evaluate the model's performance.

```
y_pred_valid = model.predict(X_valid_scaled)
print("Predictions on validation set (first 5):")
print(y_pred_valid[:5])
```

The output shows:

```
Predictions on validation set (first 5):
[31.07184262 28.54160607 36.49334721 19.73203963 41.57219194]
```

These predictions represent the model's estimates for hypertension prevalence in 2015 based on the selected features, and will be compared to the actual values in the subsequent evaluation step.

3.5 Train Linear Regression Model with Selected Features

After selecting the most relevant features, the next step was to train the **Linear Regression** model using the chosen features. The **LinearRegression** class was imported from `sklearn.linear_model` to create the model. Upon instantiation, the model was trained using the scaled training data and the target variable (2015 hypertension prevalence).

The trained model's coefficients, which represent the strength and direction of the relationship between each feature and the target, were then displayed. The coefficients and intercept values are as follows:

- **1990**: -0.7200
- **1995**: -0.2652
- **2000**: 7.9805
- **2005**: -16.8140
- **2010**: 15.3921
- **Intercept**: 37.9912

These coefficients indicate how each historical year influences the predicted 2015 hypertension prevalence. For example, the year **2010** has a positive impact (15.3921), whereas **2005** has a negative influence (-16.8140). The **intercept** (37.9912) represents the baseline hypertension prevalence when all feature values are zero.

3.6 Evaluate Linear Regression Model with Selected Features

After training the Linear Regression model, its performance was evaluated on the validation set using various evaluation metrics. These metrics include **R-squared (R^2)**, **Mean Absolute Error (MAE)**, and **Mean Squared Error (MSE)**. These are key indicators of how well the model performs in predicting the target variable (2015 hypertension prevalence).

The evaluation was carried out as follows:

```
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error

# Make predictions on the validation set
y_pred_valid = model.predict(X_valid_scaled)

# Calculate R-squared score
r2 = r2_score(y_valid, y_pred_valid)

# Calculate Mean Absolute Error (MAE)
mae = mean_absolute_error(y_valid, y_pred_valid)

# Calculate Mean Squared Error (MSE)
mse = mean_squared_error(y_valid, y_pred_valid)

# Print the evaluation metrics
print("Model Evaluation on Validation Set:")
print(f"  R-squared: {r2:.4f}")
print(f"  Mean Absolute Error (MAE): {mae:.4f}")
```

```
print(f"  Mean Squared Error (MSE): {mse:.4f}")
```

The output shows:

```
Model Evaluation on Validation Set:
  R-squared: 0.9983
  Mean Absolute Error (MAE): 0.2488
  Mean Squared Error (MSE): 0.1024
```

The results were as follows:

- **R-squared**: 0.9983
  This indicates that over 99.8% of the variance in the target variable (2015 hypertension prevalence) is explained by the model, which reflects an excellent fit.

- **Mean Absolute Error (MAE)**: 0.2488
  This value represents the average magnitude of error in the model's predictions. A lower MAE indicates that the model's predictions are close to the actual values.

- **Mean Squared Error (MSE)**: 0.1024
  The MSE penalizes larger errors more heavily. The relatively low MSE suggests that the model makes accurate predictions with minimal error.

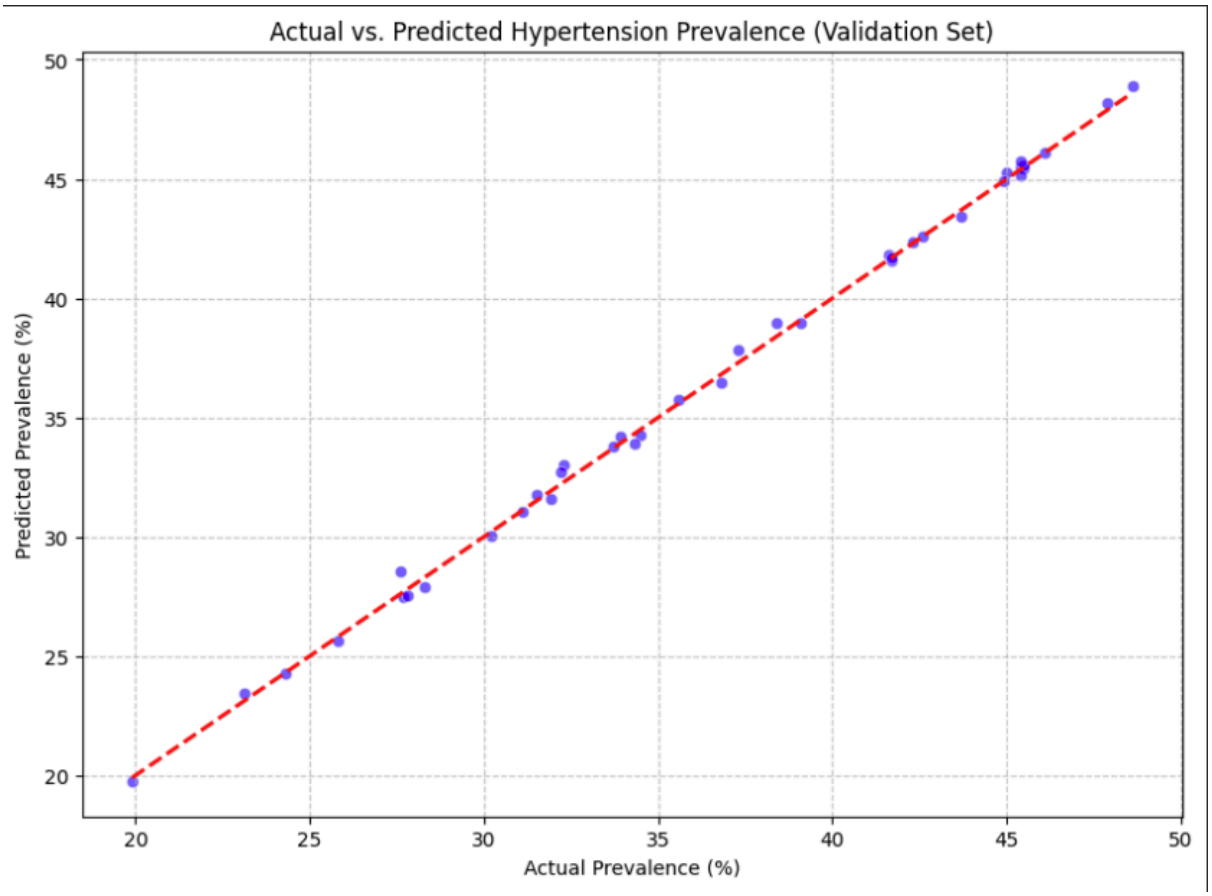
Overall, the model performed exceptionally well on the validation set, achieving high predictive accuracy and low error rates.

4. Visualization

Visualization helps us understand the model's predictions and compare them to the actual values. The following two subpoints will help assess the Linear Regression model's fit visually:

4.1 Visualizing the Regression Line

To assess the performance of the Linear Regression model, we visualize the regression line that represents the relationship between actual and predicted values. The closer the data points are to the regression line, the better the model fits the data.



The following observations can be made, which visualizes the performance of the Linear Regression model:

Close Alignment: The scatter points are very close to the red dashed regression line, which represents the "perfect prediction line" where the predicted values are equal to the actual values. This shows that the model's predictions are very close to the actual hypertension prevalence values for the validation set.

Accurate Predictions: Most of the blue points lie directly along the red dashed line, indicating that the predicted values match the actual values with minimal error. The closeness of the points to the line suggests that the model performs well on the validation set.

Linear Relationship: The regression line clearly shows a positive linear relationship between actual and predicted values. This indicates that as the actual hypertension prevalence increases, the predicted values increase similarly, confirming that the model captures the underlying trend in the data.

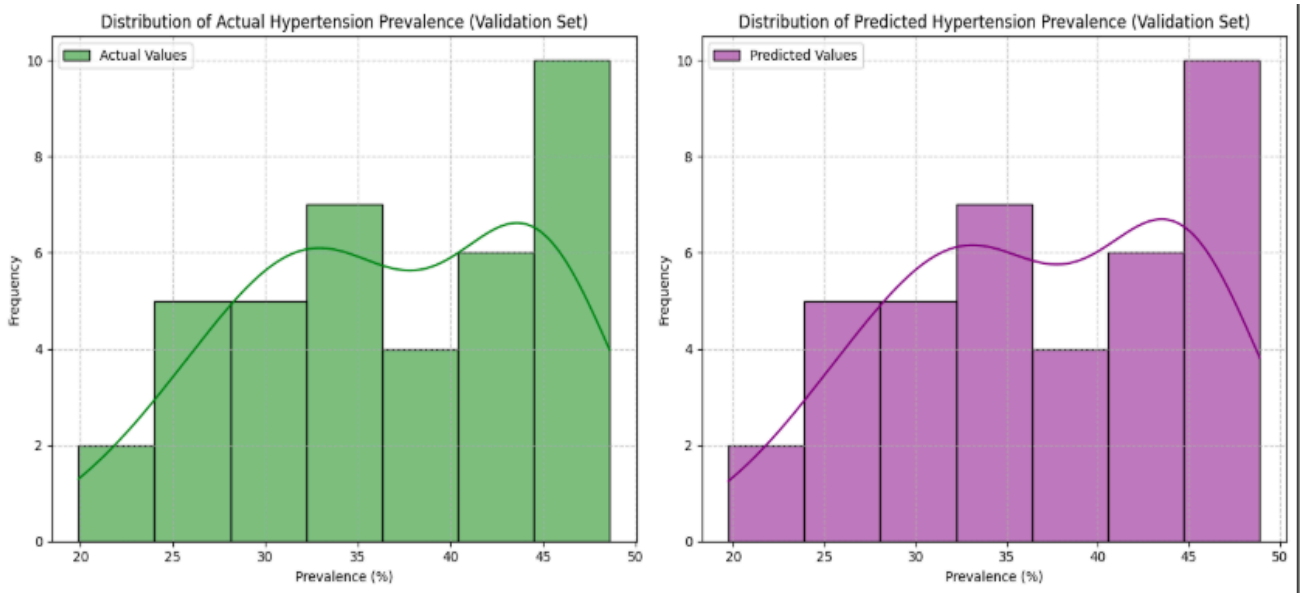
Minor Deviation: There are a few points slightly off the line, but these deviations are minimal, suggesting that the model’s error is low and predictions are reliable across the range of data.

Model Reliability: The alignment of the actual and predicted values indicates that the model is consistently providing accurate predictions, making it suitable for forecasting future hypertension prevalence.

4.2 Distribution of Actual vs. Predicted Values

To further assess the model's performance beyond point-to-point accuracy, it's insightful to compare the overall distribution of the actual validation target values (**y_valid**) with the distribution of the model's predictions (**y_pred_valid**). Ideally, these distributions should be very similar, indicating that the model not only predicts individual values well but also captures the underlying statistical properties of the target variable.

The following graph compares the distributions of the **actual** and **predicted** hypertension prevalence values for the validation set:



- **Actual Values (Green)**: The histogram on the left shows the frequency distribution of actual hypertension prevalence in the validation set. The green bars represent the actual values, and the smooth line (kernel density estimate) indicates the overall distribution.

- **Predicted Values (Purple)**: The histogram on the right shows the distribution of the predicted values by the model. The purple bars represent the predicted values, and the smooth line highlights the distribution of these predictions.

From this comparison, we can evaluate how well the model captures the overall pattern and distribution of the actual data, providing insights into its ability to generalize across the validation set.

5. Feature Selection Methods: Rationale and Observations

5.1 Observations on Model Performance and Feature Selection

After training and evaluating the model, we observed that the selected features—**1990**, **1995**, **2000**, **2005**, and **2010**—played a key role in achieving strong model performance. The evaluation metrics indicated that the model performed exceptionally well on the validation set:

- **R-squared**: 0.9983, indicating that the model explains over 99% of the variance in the target variable (2015 hypertension prevalence), demonstrating an excellent fit.
- **Mean Absolute Error (MAE)**: 0.2488, showing that the average magnitude of error between the predicted and actual values is minimal.
- **Mean Squared Error (MSE)**: 0.1024, indicating a low level of overall prediction error, with large errors penalized more heavily.

These metrics confirmed that the selected features contributed to a highly accurate model that closely matched the actual values, ensuring reliable predictions for future hypertension prevalence.

5.2 Why We Chose the Selected Features

The feature selection methods we employed—**Pearson Correlation Coefficient**, **Variance Threshold**, **Mean Absolute Difference (MAD)**, and **Mutual Information**—provided a thorough analysis of the features' relationships with the target variable.

- **Pearson Correlation Coefficient** revealed which years had the strongest linear relationships with the target (2015 hypertension prevalence), with years closer to 2015 (like 2010) showing the highest correlations.
- **Variance Threshold** helped eliminate near-constant features, ensuring that only those with sufficient variability remained in the model.
- **MAD** identified features with significant dispersion, helping to retain those with meaningful variation.
- **Mutual Information** captured non-linear relationships, ensuring that we considered complex dependencies between the features and the target.

These techniques collectively allowed us to select the most relevant and informative features for the model, improving its ability to predict hypertension prevalence accurately.

5.3 Why We Did Not Choose Certain Methods

We opted not to use some feature selection methods due to their unsuitability for the continuous nature of our dataset and regression task:

- **Chi-Square Test** is designed for categorical data and thus was not applicable for our continuous features and target variable.
- **Fisher's Score** is typically used in classification tasks to separate classes and was not suited for predicting continuous values in a regression context.
- **Dispersion Ratio** provides similar insights to **Variance Threshold** and **MAD** in terms of variability, so it was considered redundant and not included in the analysis.

By carefully selecting appropriate feature selection techniques and excluding methods that were not relevant, we ensured the model was built on the most meaningful data, leading to robust performance and accurate predictions.

6. Final Thoughts and Insights

Through our feature selection process and subsequent model evaluation, we have successfully identified the most relevant features for predicting hypertension prevalence in 2015. By leveraging techniques such as **Pearson Correlation Coefficient**, **Variance Threshold**, **Mean Absolute Difference (MAD)**, and **Mutual Information**, we ensured that the model was built on strong, informative data points. These methods enabled us to select features that not only aligned well with the target variable but also enhanced the model's ability to predict future hypertension trends.

The **Linear Regression model** showed remarkable accuracy, with a **R-squared** of 0.9983, indicating that over 99% of the variance in the target variable was captured by the model. The **Mean Absolute Error (MAE)** and **Mean Squared Error (MSE)** values further confirmed that the model provides reliable and precise predictions.

These insights indicate that our chosen features and the resulting model can serve as a robust tool for predicting hypertension prevalence, offering valuable insights into public health trends. Moving forward, the model can be further refined or applied to similar datasets, providing a foundation for informed decision-making in healthcare planning and intervention.